



# IndianOil

## JOURNEY RISK MANAGEMENT (JRM) STUDY

### Salem Terminal TO RAJA RATHNAM AGENCIES

#### Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

#### Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

#### Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

## **Compliance with the Road Transport Safety Policy (RTSP)**

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

## **Emergency Preparedness and Response**

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

## **Environmental Considerations**

The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

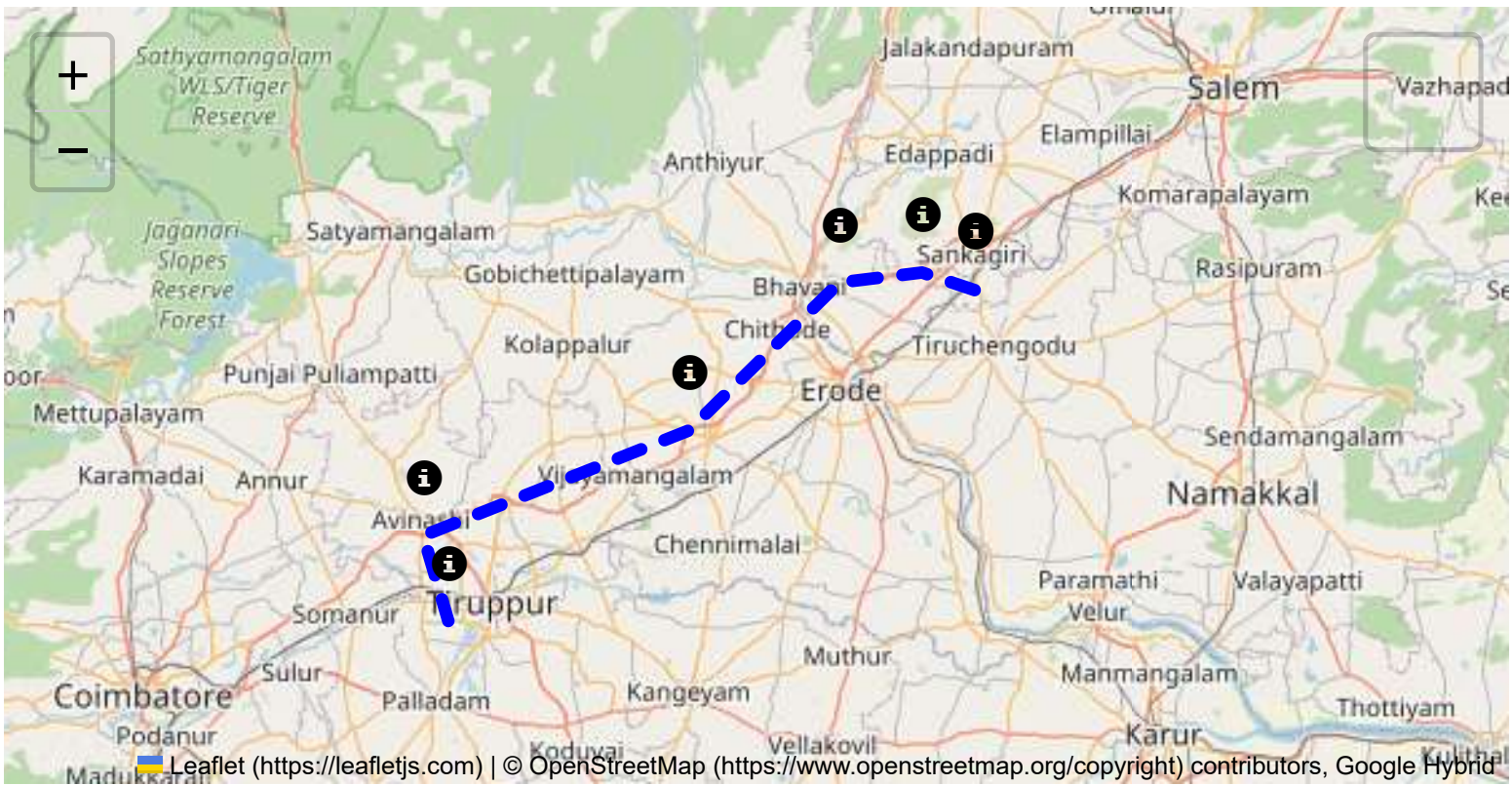
## **Journey Risk Mitigation**

The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

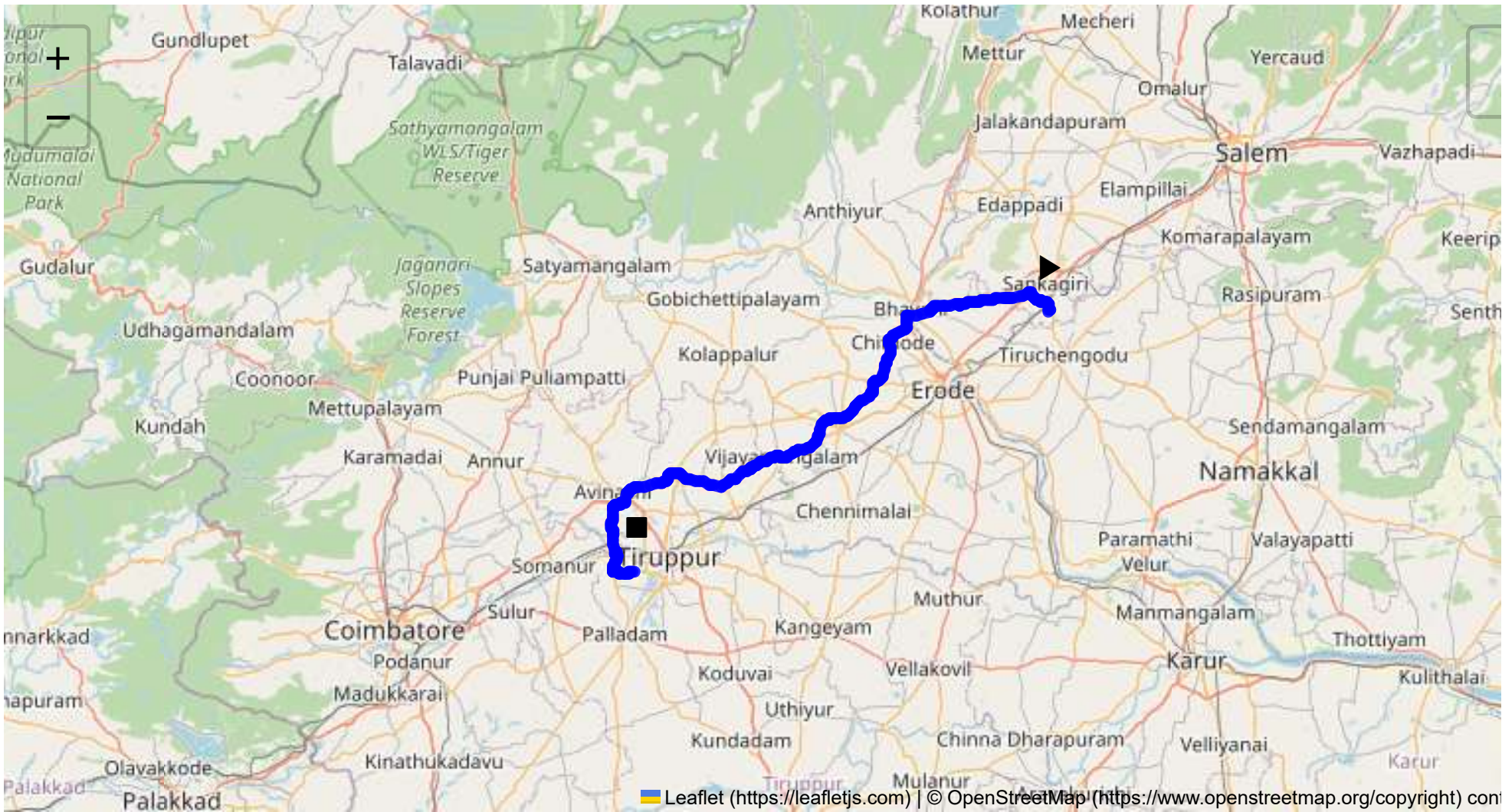
## **Compliance with Government Directives**

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.





**Route Summary:**  
**Total Distance: 100.93 km**  
**Estimated Duration: 2.0 hours**  
**Adjusted Duration (Heavy Vehicle): 2.4 hours**  
**Start: (11.4381, 77.8734)**  
**End: (11.081754, 77.29644)**



## Risk Spots

|   | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|---|--------------|--------------------|-------------|---------------------|
| 1 | Turn         | 11.43961, 77.87341 | 15 KM/Hr    | 0.08 km             |
| 2 | Turn         | 11.44029, 77.87544 | 15 KM/Hr    | 0.28 km             |
| 3 | Turn         | 11.44893, 77.87413 | 15 KM/Hr    | 1.29 km             |



|    | RiskSpotType | Coordinates        | Speed Limit | Distance from Start |
|----|--------------|--------------------|-------------|---------------------|
| 4  | Turn         | 11.45350, 77.85700 | 15 KM/Hr    | 3.28 km             |
| 5  | Turn         | 11.45845, 77.85141 | 15 KM/Hr    | 4.10 km             |
| 6  | Turn         | 11.46303, 77.84945 | 15 KM/Hr    | 4.64 km             |
| 7  | Turn         | 11.45542, 77.81725 | 15 KM/Hr    | 8.35 km             |
| 8  | Blind Spot   | 11.45634, 77.81673 | 10 KM/Hr    | 8.49 km             |
| 9  | Turn         | 11.45081, 77.77034 | 15 KM/Hr    | 13.81 km            |
| 10 | Turn         | 11.43192, 77.67803 | 15 KM/Hr    | 24.40 km            |
| 11 | Turn         | 11.18785, 77.28659 | 15 KM/Hr    | 83.57 km            |
| 0  | Roundabout   | 11.17250, 77.26863 | 15 KM/Hr    | 86.60 km            |
| 12 | Turn         | 11.14580, 77.26363 | 15 KM/Hr    | 89.65 km            |
| 13 | Turn         | 11.14029, 77.26520 | 15 KM/Hr    | 90.23 km            |
| 14 | Turn         | 11.10890, 77.27134 | 15 KM/Hr    | 93.78 km            |
| 15 | Turn         | 11.10628, 77.27042 | 15 KM/Hr    | 94.21 km            |
| 16 | Turn         | 11.10249, 77.27147 | 15 KM/Hr    | 94.66 km            |
| 17 | Turn         | 11.08036, 77.26776 | 15 KM/Hr    | 97.21 km            |
| 18 | Turn         | 11.08063, 77.27250 | 15 KM/Hr    | 97.73 km            |
| 19 | Turn         | 11.07990, 77.27587 | 15 KM/Hr    | 98.08 km            |
| 20 | Turn         | 11.07952, 77.28248 | 15 KM/Hr    | 98.85 km            |
| 21 | Turn         | 11.07878, 77.28748 | 15 KM/Hr    | 99.46 km            |

### Emergency Services

|    | type     | name                                           | coordinates            | side_of_road  |
|----|----------|------------------------------------------------|------------------------|---------------|
| 2  | hospital | Dhanvantri Multi Speciality Hospital           | 11.451362, 77.766602   | Opposite Side |
| 3  | hospital | Dhanvanthri Hospital                           | 11.4496712, 77.7593772 | Opposite Side |
| 4  | hospital | J.K.K. Trust Hospital                          | 11.4445841, 77.7307962 | Driver's Side |
| 5  | hospital | Shri Sathyanarayana Hospital                   | 11.4291297, 77.6913408 | Driver's Side |
| 8  | hospital | Thanish Siddha Hospital                        | 11.430003, 77.674964   | Opposite Side |
| 9  | clinic   | Harshitha Clinic                               | 11.4313207, 77.674718  | Opposite Side |
| 10 | hospital | Sri Kaalangi Siddhar Mooligai Vaithiya Nilayam | 11.432369, 77.674894   | Opposite Side |
| 11 | clinic   | G.K Clinic                                     | 11.4297244, 77.6749715 | Opposite Side |
| 13 | clinic   | Erode Cancer Centre                            | 11.3732, 77.649152     | Opposite Side |
| 27 | hospital | Irt Hospital                                   | 11.2803603, 77.5644118 | Driver's Side |
| 31 | hospital | Gen Siddha Hospital                            | 11.2375033, 77.5059797 | Opposite Side |

|    | type     | name                              | coordinates            | side_of_road  |
|----|----------|-----------------------------------|------------------------|---------------|
| 32 | hospital | Vijayamangalm Government Hospital | 11.2382827, 77.501687  | Opposite Side |
| 33 | hospital | Dr. N Viswanathan Hospital        | 11.2412783, 77.5005502 | Opposite Side |
| 35 | clinic   | P.M. Clinic                       | 11.2281078, 77.4647296 | Opposite Side |
| 38 | hospital | Sri Renu Hospital                 | 11.1990225, 77.4219937 | Driver's Side |

## Rest Stops

|    | type       | name                      | coordinates            | side_of_road  |
|----|------------|---------------------------|------------------------|---------------|
| 0  | fuel       | SLOA Bunk - Unit II       | 11.4543854, 77.8052665 | Driver's Side |
| 1  | restaurant | Canteen                   | 11.4478386, 77.7705441 | Driver's Side |
| 6  | fuel       | Reliance                  | 11.4325168, 77.6912865 | Opposite Side |
| 7  | fuel       | Indian Oil                | 11.4321177, 77.6765153 | Opposite Side |
| 12 | fuel       | KR Fuels - Auto LPG       | 11.4127737, 77.6747462 | Driver's Side |
| 14 | hotel      | Athithi cafe              | 11.3756993, 77.6516401 | Driver's Side |
| 15 | restaurant | Athithi Cafe              | 11.3756705, 77.6516565 | Driver's Side |
| 16 | restaurant | Canteen                   | 11.3725755, 77.6466465 | Opposite Side |
| 17 | restaurant | Kumki Restaurant          | 11.3523614, 77.644608  | Driver's Side |
| 18 | parking    | Kumki Restaurant          | 11.352414, 77.644492   | Driver's Side |
| 19 | restaurant | Paradise Biriyani         | 11.3291744, 77.6297833 | Driver's Side |
| 20 | hotel      | Hotel Annachi             | 11.3020259, 77.600017  | Opposite Side |
| 21 | hotel      | Adayar Ananda Bhavan(a2b) | 11.3018451, 77.599909  | Opposite Side |
| 22 | restaurant | Ambur Star Biriyani       | 11.3017372, 77.5999546 | Opposite Side |
| 23 | parking    | NHAI parking              | 11.3030168, 77.6009269 | Opposite Side |
| 24 | restaurant | Arukkaniyammal Mess       | 11.2982771, 77.5967424 | Opposite Side |
| 25 | fuel       | HP                        | 11.2909675, 77.5815262 | Driver's Side |
| 26 | cafe       | Backery and Cafe          | 11.2893987, 77.565338  | Opposite Side |
| 28 | fuel       | Reliance                  | 11.2575595, 77.5511245 | Driver's Side |
| 29 | fuel       | Bharat Petroleum          | 11.255444, 77.5457729  | Opposite Side |
| 30 | fuel       | Auto LPG                  | 11.2523222, 77.5400243 | Opposite Side |
| 34 | fuel       | HP Gas Station            | 11.239607, 77.4949754  | Driver's Side |
| 36 | restaurant | MDS restaurant            | 11.2058137, 77.4295619 | Opposite Side |
| 37 | restaurant | Hotel Sree Anandha Bhavan | 11.2058482, 77.4297017 | Opposite Side |

|    | type       | name                | coordinates            | side_of_road  |
|----|------------|---------------------|------------------------|---------------|
| 39 | restaurant | Shri Sai Krishna    | 11.1993912, 77.3207179 | Driver's Side |
| 40 | fuel       | HP bunk             | 11.1946674, 77.2943298 | Driver's Side |
| 41 | cafe       | Tea Shop and Bakery | 11.0859935, 77.2677    | Opposite Side |

## Welcome to the Journey Risk Management Study

### 1. Overview of the Route Map

This route extends approximately 100.93 kilometers from Sangagiri to Tiruppur via Puthur, Padaiveedu, Perundurai, and Avinashi. It follows a combination of state highways and service roads, leading primarily through rural and semi-urban areas in Tamil Nadu, India.

### 2. Typical Weather Conditions and Potential Weather-Related Hazards

The region typically experiences a tropical climate with high temperatures, particularly from March to June. Monsoon rains occur from October to December, potentially leading to waterlogging and reduced visibility. These conditions can make road surfaces slippery and increase the risk of hydroplaning.

### 3. Analysis of Traffic Patterns

- Peak Hours:** Morning and evening rush hours, usually from 8:00 AM to 10:00 AM and 5:00 PM to 8:00 PM.
- Congestion-Prone Areas:**
  - Around major junctions in Perundurai and Avinashi.
  - Bhavani Main Road in Padaiveedu may experience slow-moving traffic, particularly during market hours.

### 4. Assessment of Road Quality and Infrastructure

- Sangagiri to Puthur:** Largely in good condition but watch for occasional potholes.
- Padaiveedu to Perundurai:** Well-maintained highways with proper signage; local service roads may have uneven surfaces.
- Perundurai Bypass to Avinashi:** Generally good, but be cautious near construction zones or ongoing roadwork.
- Avinashi to Tiruppur:** Recently upgraded sections provide smooth travel, though some narrower stretches may cause delays.

### 5. Suggestions for Alternative Routes for Emergencies

- Via Erode:** From Puthur, consider diverting to Erode and then taking a southern bypass to Perundurai.

- **State Highways:** Use alternative state highways connecting smaller towns to reach Tiruppur via a less congested route.

## 6. Summary of Local Regulations Affecting Hazardous Material Transport

- Permits are required for the transport of hazardous materials. Adhere to Tamil Nadu State Pollution Control Board guidelines.
- Dangerous goods must be acknowledged at designated check posts.

## 7. Overview of Historical Incidents Involving Heavy Vehicles or Hazardous Materials

- Historical data reveals occasional accidents due to mechanical failures or driver fatigue. Incidents involving hazardous materials have been rare but can lead to significant environmental and safety concerns.

## 8. Environmental Considerations and Sensitive Areas

- Ecologically sensitive spots include agricultural zones and water bodies around Puthur and Perundurai. Minimize potential spillage risks, especially near these regions.

## 9. Analysis of Communication Coverage

- **Potential Dead Zones:** Intermittent coverage issues may arise in remote stretches near Padaiveedu and sections of the Perundurai Bypass.
- Generally strong network signals are available through more urban segments.

## 10. Estimated Emergency Response Times for Different Route Segments

- **Sangagiri to Puthur:** Approx. 20-30 minutes.
- **Padaiveedu to Perundurai:** Around 40 minutes, with longer times during peak traffic.
- **Perundurai to Tiruppur:** Emergency services from Avinashi can take around 30 minutes.

## 11. Overall Summary of Risk Assessment

This route presents moderate risk levels, with potential hazards mainly due to weather and road conditions. Traffic congestion in peri-urban stretches and construction near Perundurai and Avinashi warrant caution. Emergency plans should account for varying response times and identify secure stops along the route for potential breakdowns. Adherence to local regulations remains critical, particularly for hazardous material transport, to mitigate risks associated with traffic and environmental sensitivity. Proper preparation can ensure a safer and more efficient journey.



## Route Photos of Risky Spots



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 0.28 km

**Coordinates:** 11.44029, 77.87544



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 1.29 km

**Coordinates:** 11.44893, 77.87413





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 3.28 km

**Coordinates:** 11.45350, 77.85700



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 4.10 km

**Coordinates:** 11.45845, 77.85141





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 4.64 km**

**Coordinates: 11.46303, 77.84945**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 8.35 km**

**Coordinates: 11.45542, 77.81725**



**Risk Type: Blind Spot**

**Speed Limit: 10 KM/Hr**

**Distance from Start: 8.49 km**

**Coordinates: 11.45634, 77.81673**





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 13.81 km**

**Coordinates: 11.45081, 77.77034**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 24.40 km**

**Coordinates: 11.43192, 77.67803**





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 83.57 km**

**Coordinates: 11.18785, 77.28659**



**Risk Type: Roundabout**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 86.60 km**

**Coordinates: 11.17250, 77.26863**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 89.65 km**

**Coordinates: 11.14580, 77.26363**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 90.23 km

**Coordinates:** 11.14029, 77.26520



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 93.78 km

**Coordinates:** 11.10890, 77.27134





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 94.21 km**

**Coordinates: 11.10628, 77.27042**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 94.66 km**

**Coordinates: 11.10249, 77.27147**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 97.21 km**

**Coordinates: 11.08036, 77.26776**





**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 97.73 km

**Coordinates:** 11.08063, 77.27250



**Risk Type:** Turn

**Speed Limit:** 15 KM/Hr

**Distance from Start:** 98.08 km

**Coordinates:** 11.07990, 77.27587





**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 98.85 km**

**Coordinates: 11.07952, 77.28248**



**Risk Type: Turn**

**Speed Limit: 15 KM/Hr**

**Distance from Start: 99.46 km**

**Coordinates: 11.07878, 77.28748**

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[Download Interactive Map](#)